B3.2 Transport (Plant)

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b3-2-plant.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Transport in animals (B3.2.1–6) is in Body Systems. Image credits are on the last slide of the slides for this subtopic.

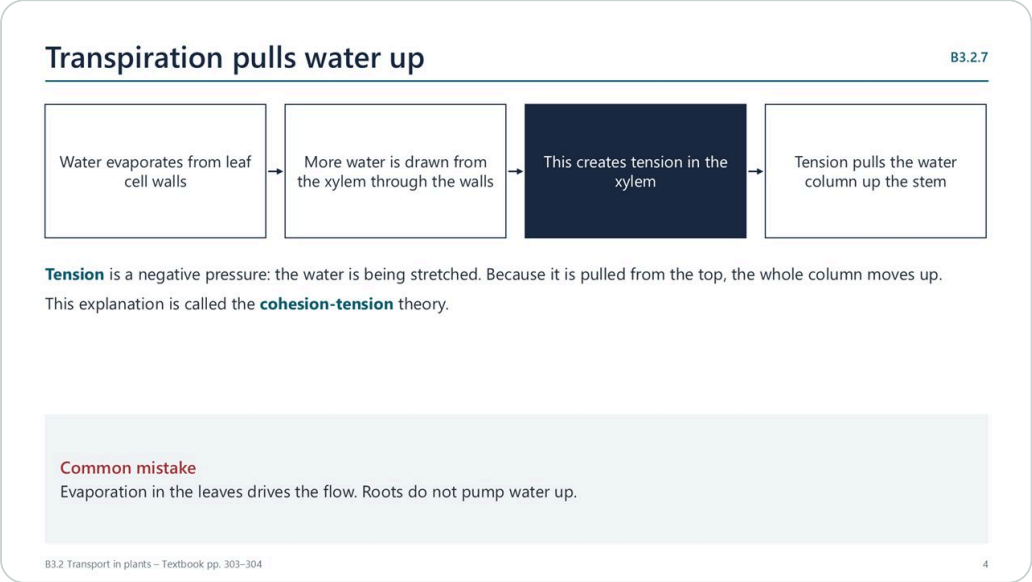
MOVING WATER

B3.2.7 How transpiration moves water from roots to leaves

pp. 303–307

1. Water evaporates from cell walls in the leaf.
2. More water is drawn out of xylem vessels and through cell walls by **capillary action**.
3. This generates **tension**, or negative pressure, in the xylem.
4. **Cohesion** between water molecules keeps the column continuous, so the tension pulls water up from the roots.

Adhesion between water and hydrophilic cell walls helps too. The plant spends no ATP on this movement, because the Sun supplies the energy for evaporation.



Evidence for tension: trunks of trees shrink slightly in the day, when transpiration is fastest. If a xylem vessel is pierced, air is drawn in and water does not spurt out.

Watch out: water is pulled from above, not pushed from below.

B3.2.8 How xylem vessels are adapted to carry water

p. 308

FEATURE	HOW IT HELPS
NO CELL CONTENTS	Unimpeded flow
INCOMPLETE OR ABSENT END WALLS	Continuous tubes, with unimpeded flow
WALLS STRENGTHENED WITH LIGNIN	Withstand tension without collapsing
PITS	Allow water to enter and leave

WHERE THE TISSUES ARE

B3.2.9 How tissues are arranged in a dicot stem

pp. 308–309

The **epidermis** forms the outer layer, and the **cortex** lies inside it. **Vascular bundles** form a ring near the outside. In each bundle, **xylem** carries water and ions towards the centre of the stem and **phloem**, which carries sugars, is on the outer side.

Tissues in a dicot stem

B3.2.9



Sunflower stem, stained section. Jon Houseman, CC BY-SA 4.0.

Tissue	Main function
Epidermis	Protects; reduces water loss
Cortex	Between epidermis and bundles; support
Vascular bundles	Ring near the outside; transport
Xylem	Inner part of each bundle: water, ions
Phloem	Outer part of each bundle: sugars

Remember
Stem: bundles in a ring near the edge, phloem outside the xylem.

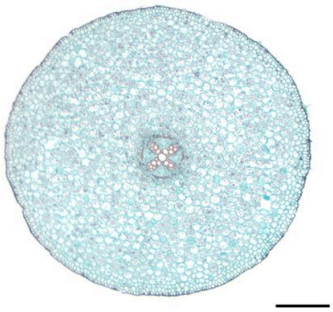
B3.2.10 How tissues are arranged in a dicot root

The **epidermis** has root hairs for absorbing water. The large **cortex** surrounds the central vascular tissue. **Xylem** forms a star or cross shape at the very centre, and **phloem** lies between its arms.

When you draw either plan diagram, show the outlines of tissues, not cells. Keep to the proportions of the micrograph, and label each boundary.

Tissues in a dicot root

B3.2.10



Buttercup (Ranunculus) root, stained section. Jon Houseman, CC BY-SA 4.0.

Tissue	Position and function
Epidermis	Outer layer; root hairs absorb water
Cortex	Wide region; stores food
Endodermis	Ring around the vascular tissue
Xylem	Star or cross at the very centre
Phloem	Between the arms of the xylem

Remember
Root: vascular tissue in the centre, xylem as a star.

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Watch out: do not confuse cortex cells with xylem and phloem. In a root, the vascular tissue is in the centre.

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